Assignment-5

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Here are the main components inside a desktop PC cabinet and their functions:

- **Motherboard:** The main circuit board that connects all PC parts together so they can communicate.
- **Processor (CPU):** The brain of the computer that processes instructions and performs calculations.
- **RAM (Random Access Memory):** High-speed temporary memory that stores active programs and data for quick access.
- **Storage (SSD / HDD):** Permanent memory used to store your operating system, software, and personal files.
- **Graphics Card (GPU):** A dedicated card that processes videos, 3D graphics, and games for your monitor display.
- **Power Supply Unit (PSU):** Converts wall outlet electricity into safe power for all internal components.
- **Cooling System (Fans / Heatsink):** Dissipates heat generated by the CPU and GPU to keep the system cool.
- **PC Cabinet / Case:** The outer frame that holds and protects all internal hardware components.

2. **Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

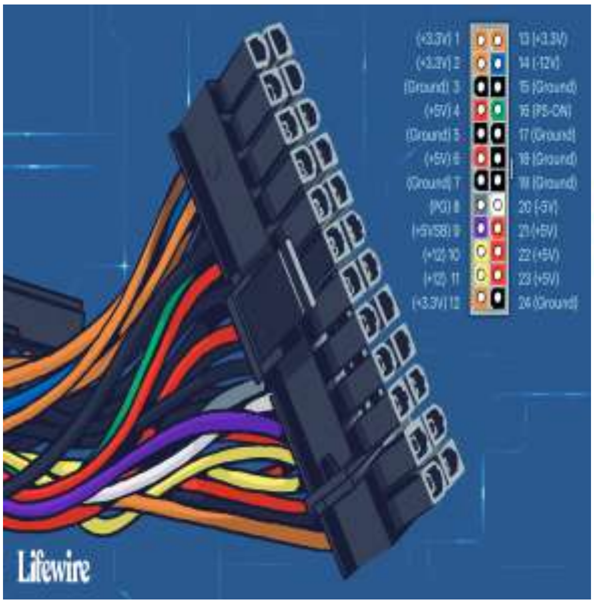
Step 1: Unplug the Power Cable

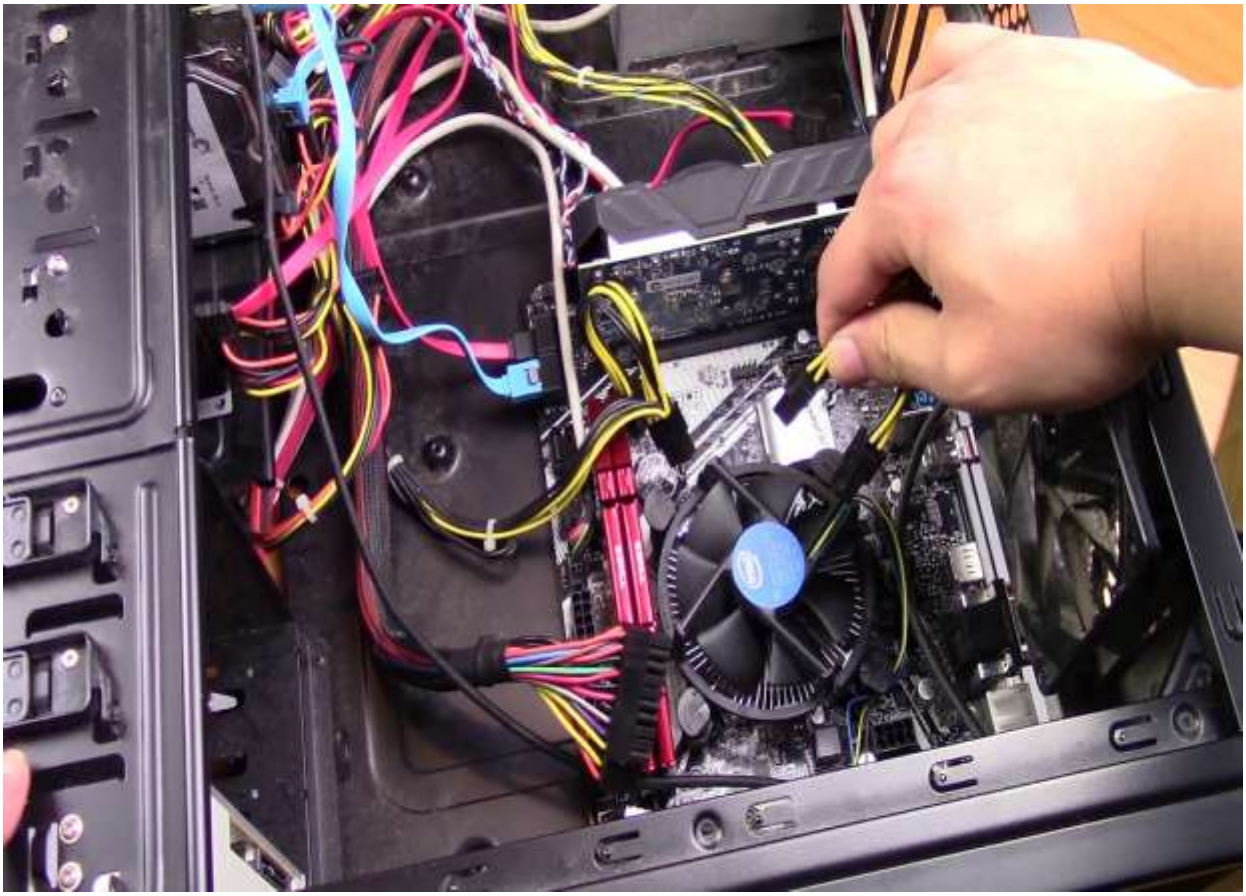
- **Description:** Turn off the power switch on the back of the PC cabinet. Pull out the main AC power cord and disconnect all external cables like the monitor, keyboard, and mouse.

Step 2: Open the Cabinet Side Panel

- **Description:** Unscrew the thumbscrews on the back panel of the computer cabinet. Slide the side metal or glass panel backward to open the cabinet.

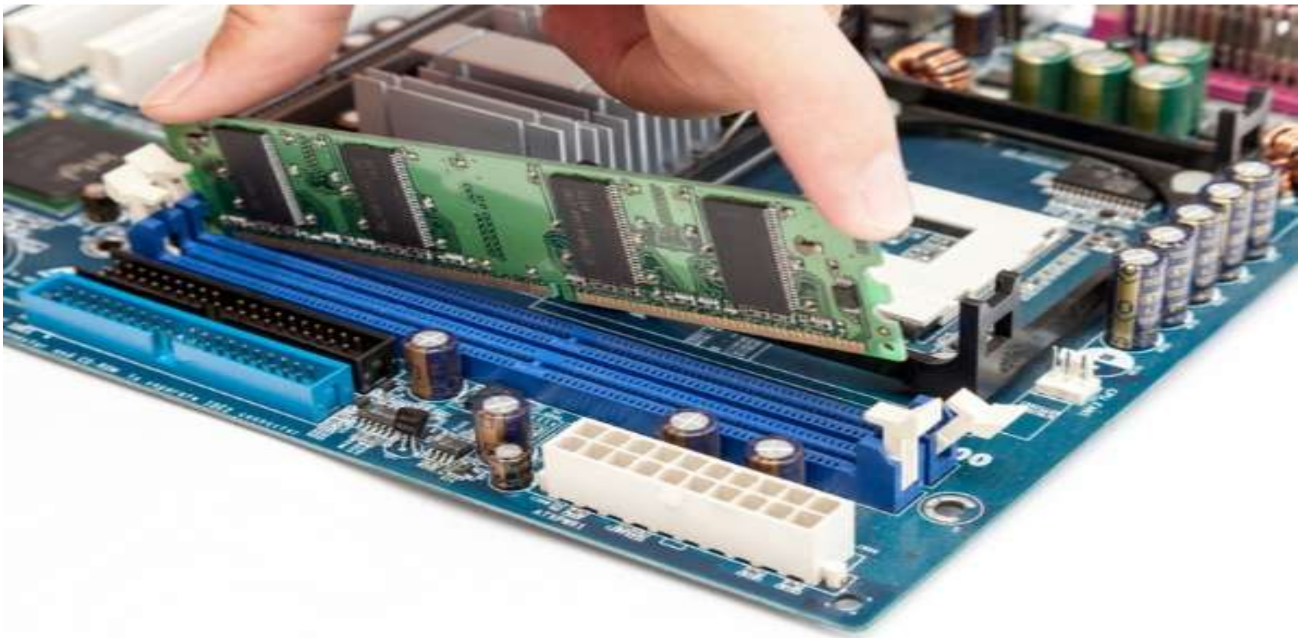
Step 3: Disconnect Internal Cables





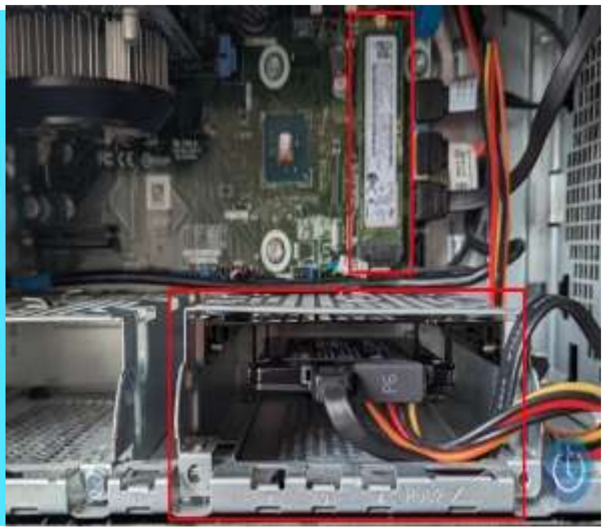
- **Description:** Unplug the main 24-pin motherboard power cable, the CPU power cable, and the GPU power cables. Disconnect the SATA cables attached to the HDD and SSD drives.

Step 4: Remove the RAM



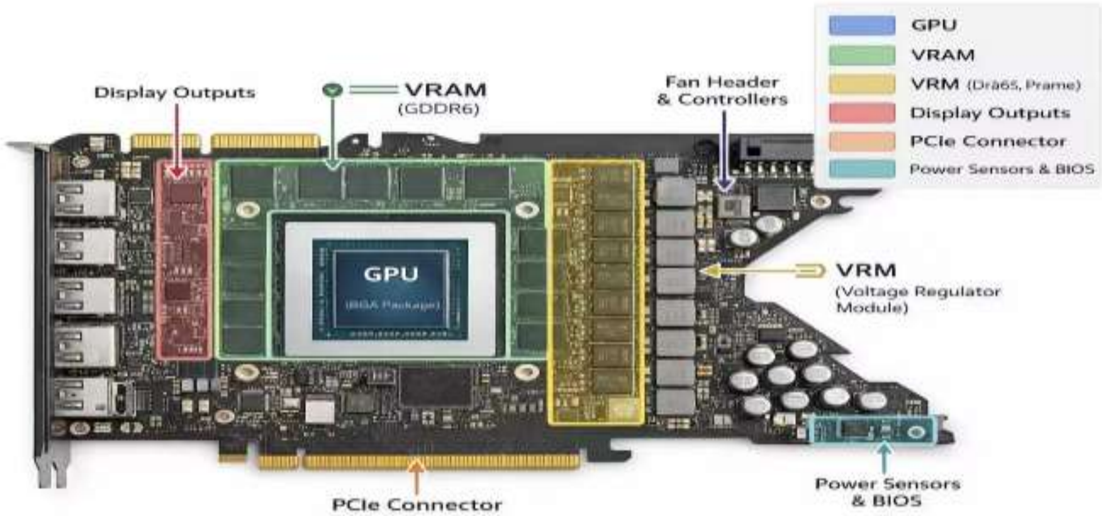
- **Description:** Push down the plastic clips at both ends of the RAM slot to unlock it. Gently pull the RAM stick straight up out of the slot.

Step 5: Remove the HDD / SSD



- **Description:** Unscrew the tiny retention screw on the motherboard and slide out the M.2 NVMe SSD. For 2.5-inch or 3.5-inch drives, unscrew them from the drive bay and remove them.

Step 6: Remove the Graphics Card (GPU)

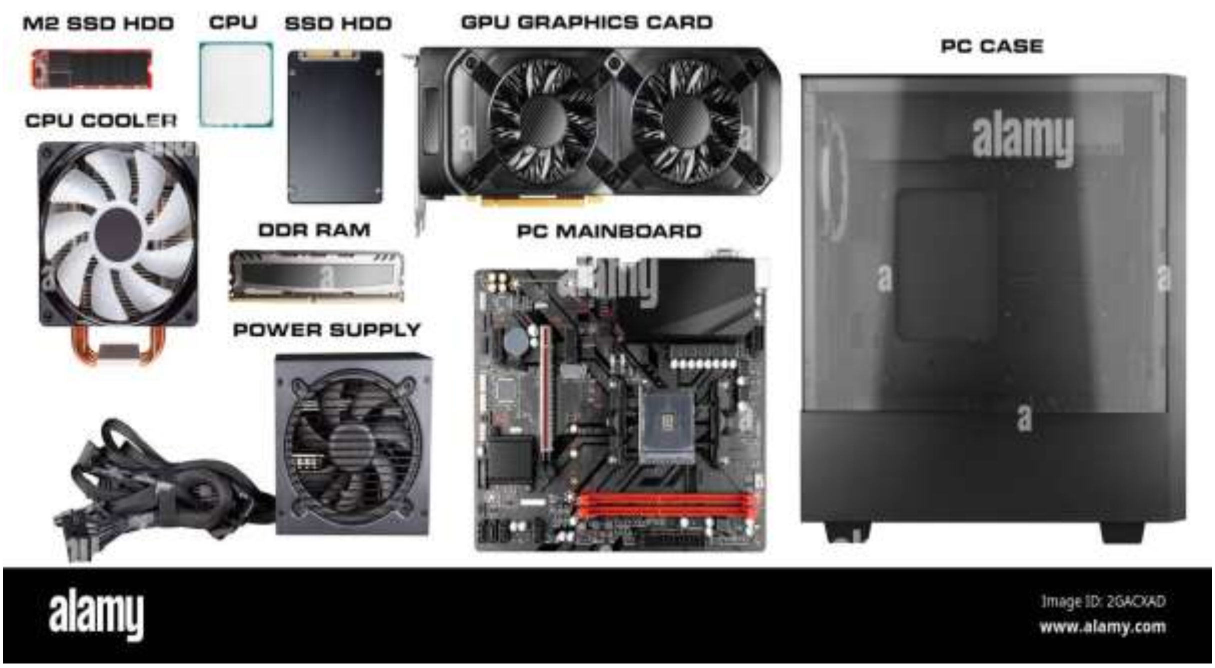


- **Description:** Remove the bracket screws holding the GPU to the cabinet frame. Press down the safety latch on the PCIe slot and carefully **pull** the graphics card straight out.

Step 7: Disassemble the Motherboard

- **Description:** Unscrew all 6 to 9 standoff screws securing the motherboard to the PC case. Hold the motherboard carefully by its edges and lift it out of the cabinet.

Step 8: All Disassembled Cabinet Parts



- **Description:** All PC parts are now completely disassembled and separated. The components include the empty cabinet, SMPS power supply, motherboard, CPU processor, RAM stick, graphics card, SSD/HDD drives, cooling fans, and connecting cables laid out on a clean flat surface.

3. **Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**

1. **Step 1: Install Motherboard:** Main board first.
    - Put the motherboard inside the case and tighten the screws.
    - **Why:** It is the main board where all parts connect.
2. **Step 2: Install CPU:** The processor.
    - Place the CPU into the socket gently and lock the lever.
    - **Why:** The CPU is the brain of the computer.
3. **Step 3: Install CPU Cooler:** Cooling fan.
    - Fix the fan over the CPU and plug its cable into the board.
    - **Why:** It keeps the CPU cool so it does not overheat.
4. **Step 4: Install RAM:** Memory stick.
    - Push the RAM stick into the slot until it clicks.
    - **Why:** RAM is needed to run apps and programs smoothly.
5. **Step 5: Install Power Supply (SMPS):** Power box.
    - Place the SMPS box in the case and tighten the screws.
    - **Why:** It supplies electricity to all computer parts.
6. **Step 6: Install HDD / SSD:** Storage drive.
    - Put the storage drive in its slot and screw it tight.
    - **Why:** It stores Windows, files, and your data.

7. Step 7: Connect Internal Cables: Main wires.

- Connect these key wires:
  - **24-pin motherboard cable** (main power)
  - **8-pin CPU cable** (processor power)
  - **SATA power & data cables** (for hard drive)
- **Why:** These wires pass power and send data between parts.

8. Step 8: Install Graphics Card (Optional): For gaming and display.

- Push the card into the PCIe slot and screw it to the case.
- **Why:** It improves display quality and gaming performance.

9. Step 9: Connect Front Panel Wires: Front buttons.

- Plug the Power button, Reset button, USB, and Audio wires into the board.
- **Why:** This makes the front case buttons and ports work.

10. Step 10: Close Cabinet: Close box.

- Slide the side panel back on and tighten the thumb screws.
- **Why:** It keeps dust out and protects internal parts.

11. Step 11: Connect External Devices: Outside connections.

- Plug in your monitor, keyboard, mouse, and power cable.
- **Why:** You need these devices to control and view the computer.

12. Step 12: Turn On PC: Final test.

- Press the power button and check if it boots up.
- **Why:** This confirms all parts are connected correctly.

4. **Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.**

Safety Steps

- Unplug the Power Cord
  - **Why:** Stops electric shock and prevents short-circuiting parts.
- Press Power Button (5 Seconds)
  - **Why:** Drains leftover electricity stored inside the PC.
- Touch Metal to Clear Static
  - **Why:** Static electricity from your body can burn sensitive chips. Touch metal to stay grounded.
- Work on a Wooden Table
  - **Why:** Carpets create static electricity. Work on a plain table.
- Hold Parts by the Edges
  - **Why:** Keeps finger oils and dirt away from gold pins and circuits.

Required Tools

- #2 Phillips Screwdriver (Magnetic)
  - **Why:** Fits almost all PC screws. A magnetic tip stops screws from falling inside.
- Small / Precision Screwdriver
  - **Why:** Needed for tiny screws used in M.2 SSDs and CPU coolers.

- Thermal Paste
    - **Why:** Fills tiny gaps between the CPU and cooler so heat travels out quickly.
- Rubbing Alcohol (90%+) & Cloth
    - **Why:** Cleans off old, dried thermal paste safely without water damage.
- Small Tray / Bowls
    - **Why:** Keeps different screws organized so you don't lose them.
- Zip Ties
    - **Why:** Binds loose wires together to keep fan blades clear and airflow smooth.

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